

plan

Implementation Plan: Workshop Curriculum Platform

Branch: 001-workshop-curriculum-platform | **Date:** 2026-08-31 | **Spec:** [spec.md](#) **Input:** Feature specification from `specs/001-workshop-curriculum-platform/spec.md`

Summary

Turn a 1.8 GB workshop recording into a browsable, searchable curriculum.

The primary requirement is a faithful, verifiable transcript (User Story 1) — everything else is built on it. On top of that sits a chapter-browsing interface modelled on the existing `ai_interviewing` module (US2), a semantic search layer over all curriculum content with stable citable passages (US3), grounded question answering that refuses rather than fabricates (US4), and a repeatable procedure for adding each new chapter (US5).

The technical approach reuses what this repository already runs — a Go backend, an Angular frontend, the existing Lumen semantic index, and the local ollama inference endpoint — and adds transcription, a passage-identity layer, and a search interface. **Five premises did not survive verification** and are corrected in Technical Context below — including two of this plan's own.

Technical Context

Language/Version: Go 1.26.2 (backend, matching ai_interviewing/platform/backend/go.mod); TypeScript/Angular (frontend, matching ai_interviewing/platform/frontend, package name ai-curriculum); Python 3.14.6 and Bash for pipeline tooling.

Primary Dependencies: ffmpeg for media handling — **but see the corrected premise below: ffprobe is NOT installed**, and ffmpeg itself resolves into an npx-managed Playwright cache. faster-whisper (CTranslate2, CPU int8, large-v3-turbo) for speech recognition — **not installed; must be added in a project-local venv**. Lumen for semantic indexing over text. ollama at <http://localhost:11434> for embeddings only — **it serves no generative model**.

Storage: Files on disk for source material, transcripts and curriculum content. SQLite for the derived index, following the pattern the existing Lumen index already uses. No external database — decision D1 makes the system single-machine.

Testing: Go's built-in testing for the backend; Karma + Jasmine for the frontend, matching the reference module; Bash harness tests for pipeline scripts, following the repository's existing convention of behavioural assertions with paired mutation proofs.

Target Platform: Linux, single machine, local/internal only (decision D1). Host container runtime is **podman + podman-compose**; docker is **absent** on this host.

Project Type: Web application (Go service + Angular client) with an offline content pipeline.

Performance Goals: Suggestions ≤ 200 ms p95 (SC-005); search results ≤ 2 s p95 (SC-006); $\geq 90\%$ top-5 accuracy on a 20-query benchmark (SC-007); $\geq 80\%$ success on queries sharing no literal words with the target (SC-008).

Constraints: CPU-only inference — the repository deliberately forces `library=cpu` and `GGML_VK_VISIBLE_DEVICES=-1` after a GPU fault corrupted the index. 8 CPUs. No content leaves the machine when a local model is configured (FR-024). No server-side CI may be added (FR-034).

Five corrected premises

These are recorded because the request or an earlier draft of this plan assumed each of them, and none held when measured. Three concern the environment and two concern this plan's own earlier claims — both kinds are listed, because a plan that hides its own corrections is not evidence of anything.

1. The reference module is not containerised — and a bespoke stack would have been a violation. The request says the curriculum must run “through the containers same way” as `ai_interviewing`. Measured: `ai_interviewing` contains **zero** container definitions; it runs as a native Go binary (`platform/bin/aicur`) serving an Angular frontend. So “the same way” cannot mean copying a setup that does not exist.

More importantly, the obvious fallback — hand-rolling a Containerfile — is forbidden. §11.4.76 (verified verbatim at `submodules/constitution/Constitution.md:7113`) mandates that for ANY containerised workload the project **MUST** consume `git@github.com:vasic-digital/containers.git` as a git submodule, and clause (4) requires **extending it, never reimplementing it**. It was not declared in this tree.

Resolution: the submodule is now mounted at `submodules/containers` — the tree's `submodules/` path is for non-product infrastructure gitlinks (`constitution`, `superspec`), and `containers` is a consumed Go library, not a product. It classifies OWNED automatically, because `github.com/vasic-digital` now hosts three declared submodules and the fleet roster derives ownership from that evidence rather than a hardcoded list.

What it actually provides — verified by building it and running its tests, not read off its README: six runtime backends behind one interface with **podman first, explicitly for rootless** (satisfying §11.4.161); compose orchestration that detects the compose CLI and resolves to `podman-compose` on this host, correctly suppressing the docker-only `--wait` flag and defending against the `podman-docker` shim; TCP/HTTP/gRPC health checkers; lazy boot with idle shutdown; and a `cmd/boot` CLI. It is Go-only, which fits — the workshop backend is Go 1.26.2.

Three gaps named rather than discovered later: - **No bash boot wrapper exists.** The request's “all mandatory bash scripts” is not satisfied by the submodule; lifecycle is a Go API plus Go CLIs. ~~Thin~~

workshop wrappers that invoke `cmd/boot` are *consumption*, so they are §11.4.76-legal. Anything beyond thin wrapping must be contributed upstream.

Corrected 2026-09-01 — `cmd/boot` is not usable as that entry point.

This bullet named the wrong one, and the error was not cosmetic: it would have specified an instrument that reports success over an absent system. Measured in `submodules/containers/cmd/boot/main.go` at gitlink 4dab992 — its flags are only `--env`, `--project`, `--timeout`, `--help`, so there are **no lifecycle verbs and no compose-file flag**; it hardcodes `{"helixagent": localhost:7061}` (`main.go:148–153`); its `BootManager` is built with **no orchestrator and no health checker**; and `runBoot` returns `nil`, so it **exits 0 having started nothing**. A wrapper trusting that exit code would report a healthy stack that does not exist — the §11.4.76(5) bluff shape exactly. *Resolution*: the control plane consumes the submodule's **Go API** (`pkg/compose`, `pkg/health`, `pkg/runtime`) through a `workshop-boot` adapter in `workshop/platform/orchestration/`, which is what §11.4.76(3) actually prescribes. Still consumption — the adapter contains no runtime detection, `compose` driving or health-check logic of its own. `cmd/boot`'s four defects are upstream debt to be fixed there per §11.4.76(4), staged in `workshop/platform/upstream-contributions/`. - `DefaultHelixServices()` is `postgres/redis/etcd`-shaped, irrelevant here — but not a defect, since `ComposeProject.File` accepts an arbitrary `compose` file, so the workshop supplies its own. - **`submodules/containers/scripts/build-all.sh` is broken upstream**: it builds `./cmd/core/` and `./cmd/host-agent/`, neither of which exists. Do not use it as a model.

One blocker it exposed — now CLEARED; recorded rather than deleted. The submodule's own four carriers are 414/246/24/24 lines — four genuinely different documents, a §11.4.157(B) violation latent upstream. The new C8 in-submodule lockstep check catches it, **which currently blocks the pre-push hook**. Being fixed upstream per §11.4.76(4), not worked around locally.

RESOLVED 2026-09-01. The finding was correct; only its tense is now wrong, and a plan that leaves a cleared blocker described as live is as misleading as one that hides a real one. The fix went upstream — not around it — exactly as §11.4.76(4) requires:

- **Upstream commit** 4dab992 “*govern(carriers): restore §11.4.157(B) four-carrier lockstep*”, touching all four carriers (+2456 / -352). Pushed, not merely local.

- **Gitlink bumped and verified:** `git ls-tree HEAD submodules/containers → 160000 commit 4dab992582666a64a4353cd593704cdc969aaa1e`. The umbrella consumes the fixed commit, so the resolution is pinned rather than incidental to a checkout.
- **Measured carrier sizes now:** 703 / 703 / 703 / 703 lines, replacing 414 / 246 / 24 / 24.
- **The instrument was re-run, not assumed** (per §11.4.6 — a fix is not a pass until the gate that caught it is observed green). `bash scripts/verify-governance-cascade.sh` on 2026-09-01:

```
PASS  C8 IN-SUBMODULE-LOCKSTEP – all 7 owned submodule(s)
carry four agent carriers with
      byte-identical bodies once the per-agent header is
normalised (§11.4.157(B))
CM-GOVERNANCE-CASCADE: 12 PASS, 0 FAIL, 0 ENV, 7 NOTE
rc=0
```

The pre-push hook is therefore no longer blocked by this, and no task is needed to clear it. Note the boundary this claim does and does not cross: what was measured is the governance cascade returning `rc=0` with C8 green. That is the check that was blocking. It is not a claim that every other pre-push gate passes — none of those were run for this correction, and their state is not asserted here.

2. Generative capability — resolved, and the answer is “none”.

Superseded by premise 5 below, which measured it. US1–US3 are deliberately sequenced to deliver value without it, and FR-025 already requires browsing and search to work when answering is unavailable, so US4 degrades to a documented operator step rather than blocking the feature.

3. `ffprobe` is not installed, and `ffmpeg` is not durably installed either.

An earlier draft of this plan asserted “`ffmpeg/ffprobe 7.0.2` (present on host)”. Measured: `$(command -v ffprobe)` resolves to a **symlink to Playwright’s `ffmpeg` binary** and rejects `-show_format` (“Unrecognized option”). The original probe ran `ffprobe --version`, which the `ffmpeg` binary accepts — so a tool that does not exist looked present. Both `ffmpeg` and `ffprobe` resolve into `~/ .cache/ms-playwright/ffmpeg-1011/`, an `npx`-managed cache that a `npx playwright uninstall` or a cache clean would remove.

Consequence: any pipeline step invoking `ffprobe` fails here with what reads like a bad-argument error rather than a missing-binary error. The pipeline **MUST** detect its media tooling rather than assume it (Environment Adaptability), and **MUST NOT** depend on a browser-automation cache for a production content pipeline.

4. No speech-recognition engine is installed. `/usr/bin/whisper` is `whisper-1.3.1-alt1`, a microphone-loopback GUI by a different author — not OpenAI Whisper; `import whisper` fails. ollama's `/v1/audio/transcriptions` endpoint is real, but its Whisper strings are an *encoder* projecting audio into an LLM token space; `word_timestamps`, `avg_logprob` and `no_speech_prob` appear nowhere in the binary, so it cannot satisfy FR-002 (timestamps) or FR-003 (confidence).

5. There is no generative model, so User Story 4 cannot run at all today. ollama serves exactly two models and they are the **same** embedding model (`jina-bert-v2`, 160.28M, an encoder with no LM head). The entire model store is 309 MB across 7 blobs — too small to hold a decoder. US4 is blocked on an operator `ollama pull`, not on code. Corroborating trap: `~/helix-agent` is a 23.7 MB **cluster mesh node** (SWIM / WireGuard / etcd), not an LLM agent; and HelixLLM appears in this repository only as portfolio marketing copy.

Measured recording properties (these, not “1.8 GB”, drive every estimate): duration **01:55:28.75 (6928.75 s)**, H.264 1920x1080 24 fps, AAC-LC 48 kHz stereo — but the stereo is **dual-mono** (L-R difference -90.3 dB), so channel-based speaker separation is impossible. Uniformly AGC-compressed. Only 8 long silences totalling 41.33 s (0.597%). The notes PDF has a real text layer (3,034 words) — a **summary**, roughly 26 words per minute of recording, and demonstrably erroneous (“Spatkit” for SpecKit), which is why the spec makes the recording authoritative.

Open items entering Phase 1:

- Whether Lumen's chunk identifiers are stable across re-index — determines whether FR-037 needs its own identity layer. Highest-risk unknown: SC-016 fails silently if wrong. Research pending.
- Whether the `cp314` wheels for `CTranslate2/faster-whisper` install and run on this host.
- Real transcription throughput, settled by a 5-minute calibration run on an already-extracted 300 s sample.

Constitution Check

GATE: Must pass before proceeding. Re-check after design phase.

Principles taken from `.specify/memory/constitution.md`, which inherits from the constitution submodule.

Principle	Status	Notes
Evidence-Based Claims — no guessing, no fabrication (§11.4)	PASS	The spec's Context table records only measured values. FR-004 requires a <i>measured</i> accuracy figure, not an assertion. FR-033 forbids reporting a check as passed when it could not run. This plan corrects two premises that measurement disproved rather than carrying them forward. Resolved 2026-09-01, after this row was first written. workshop carried none of the four carriers (coverage 20/24) and verify-governance-cascade.sh FAILED C1 and C6 on it. All four are now present and committed in that submodule at 55076bf — committing mattered, because
Governance Fidelity — four carriers in lockstep; constitution is authoritative	PASS (was NEEDS ATTENTION)	

Principle	Status	Notes
		until then a fresh git submodule update --init would have reconstructed a workshop without them. Coverage is 24/24; the verifier reports 10 PASS / 0 FAIL / rc=0, and its --prove-failure — which was itself inoperative, failing at CONTROL so zero mutations were ever reached — now runs the real entry point first. FR-032 and SC-012 require every check to carry a paired proof that it fails when its condition is broken. The transcription pipeline reads source media and never writes to it (FR-006). FR-030/031 mandate the doc set including an explicit statement of what the system cannot do. FR-035 covers CONTINUATION.md. Execution Strategy below marks the correctness-critical components [TDD]. Transcription is
Isolation by Default — mutation-paired gates; hardlinked backup before destructive ops (§1.1, §9)	PASS	
Comprehensive Documentation — CONTINUATION.md updated; honest boundaries	PASS	
Quality Over Speed — 60% RAM cap, TDD, lint/typecheck before “done”	PASS	

Principle	Status	Notes
		chunked and resumable (FR-029), keeping memory bounded rather than loading a 1.8 GB file whole.
§11.4.156 — no active server-side CI	PASS	FR-034 and SC-014 forbid adding any. The repository's local pre-push gate remains the enforcement point.

Gate result (initial): PROCEED, with one blocking prerequisite — workshop governance onboarding. That prerequisite is now **discharged** (see the row above).

Post-design re-check (2026-09-01, after Phase 1)

Re-evaluated against the design rather than the intent, as the gate requires.

Principle	Status	What the design does about it
Evidence-Based Claims	PASS	Phase 0 replaced five assumed premises with measurements and withdrew two of this plan's own claims explicitly. <code>research.md</code> carries an UNVERIFIED register (U1–U5) naming what settles each.
Honest Instruments	PASS	The 0/1/2 contract is bound into the design, not bolted on: <code>Answer.status</code> has unavailable as a first-class third state distinct from

Principle	Status	What the design does about it
		declined, and D-SEARCH-4 forbids deriving degradation from a liveness probe — measured necessity, since health_check returned OK 2 ms before semantic_search failed.
Governance Fidelity	PASS	24/24 carriers, committed.
Isolation by Default	PASS	Every gate this feature adds owes a paired mutation proof (FR-032/SC-012). The quickstart enumerates 14 such proofs as owed rather than assumed.
Comprehensive Documentation	PASS	FR-030/031 require the doc set to state what the system cannot do — which now includes real limits: media is reachable only via transcripts (D-SEARCH-3), and answering is asynchronous, not instant (D-LLM-5).
Environment Adaptability	PASS	The pipeline must detect its media tooling rather than assume it — a rule this plan earned by asserting fprobe was

Principle	Status	What the design does about it
Quality Over Speed	PASS	present when it is a Playwright symlink. Transcription is chunked and resumable, so memory stays bounded rather than loading a 1.8 GB file whole.
§11.4.156 — no server-side CI	PASS, and strengthened	Enforcement itself was defective: gate E swept only the umbrella, so an added workflow inside workshop/ would have shipped unnoticed. Closed and mutation-proven on 2026-09-01.

Post-design gate result: PROCEED to /speckit-tasks. No unjustified violation. The two Complexity Tracking entries stand, and the second is no longer speculative — D-SEARCH-1 measured that Lumen chunk IDs are content-derived, so the passage-identity layer is required.

Project Structure

Documentation (this feature)

```

specs/001-workshop-curriculum-platform/
├─ spec.md                # Feature specification
├─ plan.md                # This file
├─ research.md            # Phase 0 consolidated findings
├─ research/
│   └─ transcription.md   # ASR engine, accuracy methodology,
resumability
│   └─ search-architecture.md # Index design, passage identity,
autocomplete
│   └─ llm-bridging.md    # Provider seam, grounding, refusal
design
├─ data-model.md          # Phase 1 entities
├─ contracts/             # Phase 1 interface contracts

```

```

├─ quickstart.md          # Phase 1 validation guide
├─ tasks.md               # Task breakdown
├─ checklists/
  └─ requirements.md      # Spec quality checklist

```

Source Code (repository root)

```

workshop/                  # the feature's home (a
submodule)
├─ CLAUDE.md AGENTS.md QWEN.md GEMINI.md  # governance carriers
- PREREQUISITE, absent today
├─ chapters/
  └─ 01/                   # existing: recording
parts, .sha256, notes PDF
  └─ transcript/          # NEW: machine transcript +
corrections + evidence
├─ curriculum/           # NEW: published, browsable
chapter content
├─ platform/             # NEW: mirrors
ai_interviewing/platform layout
  └─ └─ backend/          # Go service: content,
search, answering
  └─ └─ frontend/        # Angular client: browse,
watch, search UI
  └─ └─ bin/
├─ pipeline/             # NEW: offline content
pipeline
  └─ └─ transcribe/       # media → timestamped
passages
  └─ └─ ingest/           # passages → stable ids →
index
  └─ └─ crossref/        # passage relationships
├─ scripts/              # existing archive/extract/
hooks + NEW run/build/test
  └─                      # NO containers/ dir -
orchestration is CONSUMED from
  └─                      # submodules/containers; a
local stack violates §11.4.76(4)
  └─ docs/                # NEW: quickstart, user
guide, manual, FAQ, training

```

Decoupling (operator directive, 2026-09-01): every component and service must be fully decoupled and reusable (FR-043–FR-045, SC-019). Reusable units live in pkg/, never internal/ — Go’s internal/ is importable only from within its own module, so it forecloses reuse by language rule rather than by style. Only internal/store stays private, because chapter and curriculum persistence is the one genuinely workshop-specific unit. Where reuse is real, a unit becomes its own module on the digital.vasic.containers pattern.

Structure Decision: mirror `ai_interviewing/platform's backend/ + frontend/ + bin/ split`, because the spec's FR-013 requires a person familiar with one module to navigate the other, and because that layout is already proven in this repository. Everything the workshop needs that `ai_interviewing` does not have — the transcription pipeline, passage identity, containers — is added alongside rather than by reshaping the reference.

The feature lives inside the `workshop/ submodule`, not the umbrella. That keeps the curriculum independently cloneable and keeps the umbrella's working tree clean, which FR-036 requires.

Execution Strategy

TDD Requirements

Marked [TDD] in the task breakdown. Chosen where a defect would be silent rather than loud.



Passage identity and cross-reference resolution: SC-016 demands 100% of references to passages the curriculum **owns** — transcripts and its own documentation — still resolve after correction plus re-index. This fails *invisibly* — links keep rendering, they just point at the wrong text. Tests must exist before the implementation.



Code-passage identity (SC-016a): for source code the curriculum does **not** own, no anchor can be written into the file, so identity is keyed on symbol path plus a rename alias table (`symbol_aliases`). The criterion is deliberately weaker and is **100% of stale code references fail loudly** — never a silent re-point at different code. That guarantee is exactly the kind a naive implementation converts into a confident wrong answer, so its tests also land first. Contract: `contracts/passage-contract.md` §6.2–6.6, §8 R3/R4, gate G-PID-4. Carries the open **P-U1**: the symbol matching key has no confirmed producer yet.



Refusal behaviour in question answering: SC-010 requires 10/10 unanswerable questions declined. A system that fabricates confidently is the single worst outcome this feature can produce, and the governing constitution names it as a release blocker.



Citation verification: SC-009 requires 100% of citations to genuinely support their claim. Attaching a citation is easy; verifying it supports the claim is the actual work.



Three-state search contract: results / no-match / cannot-answer. This repository has already shipped the failure once — a saturated backend produced timeouts that a naive client would render as “no results”.



Transcript accuracy measurement: the sampling and scoring harness must be correct before any accuracy figure is published, or the number means nothing.



Extension procedure idempotency: FR-027 requires a second run to change nothing.

Parallel Execution Opportunities

Marked [SUBAGENT]. These share no files.



Transcription pipeline and **frontend search UI** — different languages, different directories, joined only by the passage contract.



Documentation set (quickstart, user guide, manual, FAQ, training) can proceed once the contracts are fixed, independently of implementation.



Container definitions and **backend service** — the container work depends only on the run commands, not on their internals.



workshop governance onboarding is fully independent of all feature work and should start immediately; it is a prerequisite for completion, not for progress.

Human Checkpoints

The agent pauses and waits for explicit approval at each.

1. **After Phase 0 research** — before committing to an ASR engine and an index architecture, because both are expensive to reverse.
2. **After foundational setup** — workshop governance onboarding green, project structure in place, gates passing.

3. **After User Story 1** — the transcript and its *measured* accuracy figure reviewed. This is the gate that matters most: everything downstream inherits this artifact's quality.
4. **After each subsequent user story** — behaviour checked against that story's acceptance scenarios.
5. **Before merge** — full gate suite green, CONTINUATION.md synced, all submodules and the umbrella committed and pushed clean (FR-036, SC-015).

Review Gates

Marked [REVIEW].

☐

Passage contract (identity, timestamps, provenance) — review before anything consumes it; every other component depends on its shape.

☐

Privacy handling — the redaction path (FR-039) and the local-only guarantee (FR-024) concern an identifiable third party. Review before integration.

☐

Answering prompt and refusal thresholds — review before enabling, because a permissive threshold silently converts refusals into fabrications.

☐

Container definitions — first containers in this module; review before they become the pattern others copy.

Complexity Tracking

Violation	Why Needed	Simpler Alternative Rejected Because
Adding a submodule dependency (vasic-digital/containers) that the reference module	Not actually optional: §11.4.76 mandates it for ANY containerised workload and clause (4) forbids	Running natively like ai_interviewing adds zero new concepts — but forgoes containers entirely, which the operator asked for explicitly, and inherits the host-drift problem this repo has repeatedly been bitten by (podman-not-docker, ugrep-not-GNU-grep, a GPU that must stay out of the

Violation	Why Needed	Simpler Alternative Rejected Because
does not use	reimplementing it. Hand-rolling a Containerfile would satisfy the operator's container request while violating the constitution the same request demands be respected.	inference path). Writing a bespoke Containerfile was rejected outright as a \$11.4.76 violation, not merely as inferior.
A passage-identity layer duplicating what the index appears to provide	FR-037 requires identifiers that survive re-indexing AND text correction.	Reusing Lumen's chunk identifiers was the preferred outcome and is now rejected on measurement (D-SEARCH-1): a controlled three-run re-index showed a one-character typo fix changes the id, while a line shift does not — they are content-derived, exactly one of the two forms FR-037 forbids, and the one that breaks on the operation this feature performs most. Built on them, SC-016 would score 0% on every corrected passage with no visible symptom. The ids are not exposed through the CLI or MCP either, leaving only a positional key, which the same experiment showed moves.

Phase Status

- **Phase 0 (Research):** IN PROGRESS. Three research streams dispatched — transcription, search architecture, LLM bridging. Consolidated into `research.md` on completion.
- **Phase 1 (Design & Contracts):** NOT STARTED. Requires Phase 0.